EE 316

Section 3

Pre Lab 07

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Characteristics of BJT and Amplification Behavior

Introduction:

This laboratory introduces the basic concepts and characteristics of bipolar junction transistors (BJTs), focusing on NPN transistors in a common-collector configuration. The experiment examines the relationship between base current, collector current, and collector-emitter voltage to better understand transistor behavior and operating characteristics.

Simulations and Results:

The circuit in Figure 1 was built and simulated in LTSpice. Figure 2 shows the LTSpice schematic. To get the simulation results in Table 1, V1 was kept constant at four different voltage values (4V, 6V, 8V, and 10V) while V2 was swept from 0-10V in increments of 0.1V. The characteristic curves in Figure 3 were then graphed using the V_{CE} and I_C values in Table 1. Table 2 shows the Q-point behavior for each characteristic curve and the midpoint biasing Q-point.

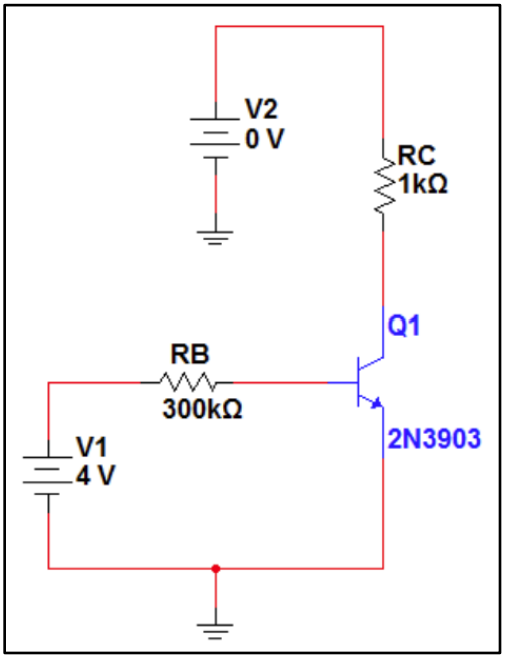


Figure #1: Common Collector Circuit Schematic (Lab Manual)

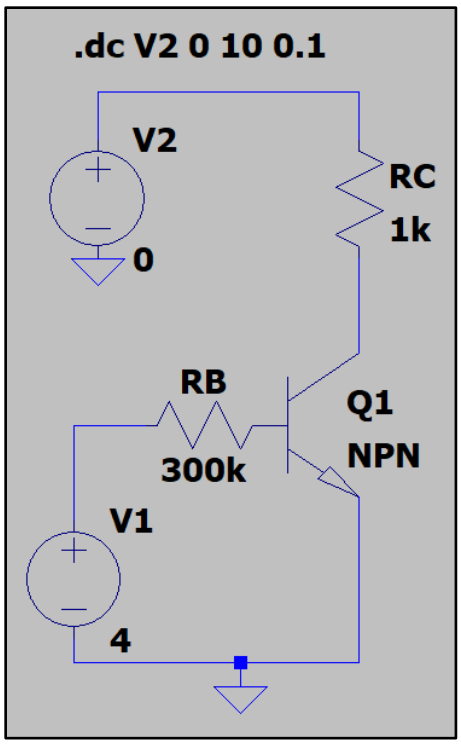


Figure #2: LTSpice Schematic (Common Collector Circuit)

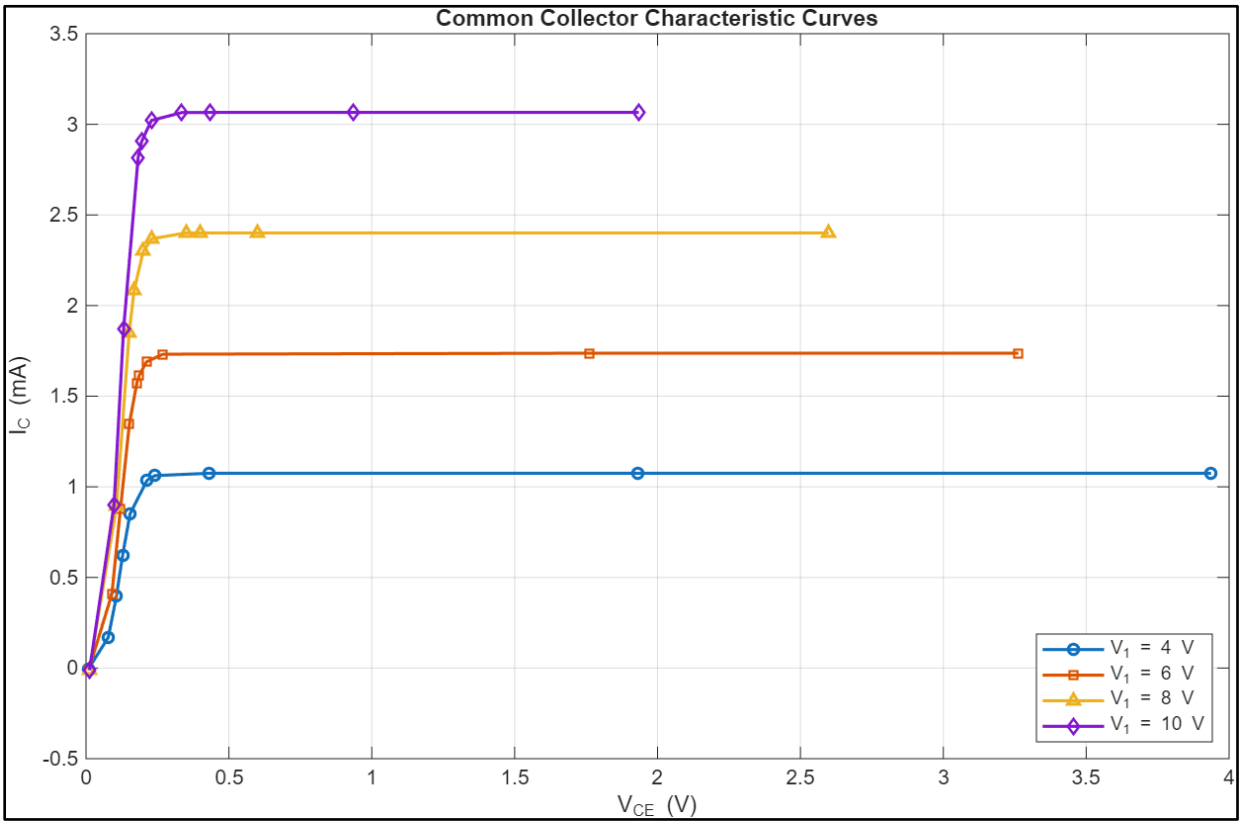


Figure #3: Common Collector Characteristic Curves

V_1 (V)	V_2 (V)	I_B (µA)	I_C (mA)	I_E (mA)	V_{CE} (V)	β
4	0	11.117079	-0.007359	-0.0037571	0.0073599	-0.000662
	0.25	10.899143	0.169282	-0.1801830	0.0778308	0.015532
	0.5	10.829636	0.395721	-0.4065514	0.1062680	0.036541
	0.75	10.792602	0.621101	-0.6318938	0.1279331	0.057549
	1	10.76553	0.849843	-0.8606096	0.1539531	0.078941
	1.25	10.749063	1.0369467	-1.0476961	0.2139410	0.096469
	1.3	10.747213	1.0617948	-1.0745447	0.2410873	0.098797
	1.5	10.746357	1.0746281	-1.0853745	0.4311629	0.099999
	3	10.746357	1.0746357	-1.0853821	1.9292254	0.100000
	5	10.746357	1.0746359	-1.0853822	3.936947	0.100000
6	0	17.7361000	-0.0096130	-0.0081231	0.0096130	-0.000542
	0.5	17.4909000	0.4094280	-0.4269180	0.0905744	0.023408
	1	17.4287000	0.8794090	-0.8968370	0.1205910	0.050458
	1.5	17.3930000	1.3482200	-1.3656100	0.1517810	0.077515
	1.75	17.3800000	1.5721500	-1.5895300	0.1778330	0.090457
	1.8	17.3778000	1.6139300	-1.6313100	0.1860300	0.092873
	1.9	17.3740000	1.6886600	-1.7060400	0.2113360	0.097195
	2	17.3719000	1.7316700	-1.7490400	0.2683260	0.099682
	3.5	17.3716000	1.7371600	-1.7545300	1.7628400	0.100000
	5	17.3716000	1.7371600	-1.7545300	3.2628400	0.100000
8	0	24.3704000	-0.0111156	-0.0132548	0.0111156	-0.000456
	1	24.0928000	0.8932120	-0.9173050	0.1067880	0.037074
	2	24.0324000	1.8490600	-1.8730900	0.1509420	0.076940
	2.25	24.0224000	2.0818900	-2.1059100	0.1681090	0.086665
	2.5	24.0140000	2.2996900	-2.3237100	0.2002450	0.095765
	2.6	24.0115000	2.3690900	-2.3931000	0.2309130	0.098665
	2.75	24.0104000	2.4007000	-2.4247200	0.3492820	0.099986
	2.8	24.0104000	2.4009900	-2.4250000	0.3990110	0.099998
	3	24.0104000	2.4010400	-2.4250500	0.5989620	0.100000
	5	24.0104000	2.4010400	-2.4250500	2.5989600	0.100000
10	0	31.0128000	-0.0121750	-0.0188378	0.0121750	-0.000393
	1	30.7575000	0.9019340	-0.9326920	0.0980658	0.029324
	2	30.6976000	1.8686200	-1.8993200	0.1313790	0.060872
	3	30.6631000	2.8175500	-2.8482200	0.1824420	0.091887
	3.1	30.6605000	2.9054600	-2.9361200	0.1945240	0.094762
	3.25	30.6572000	3.0212500	-3.0519100	0.2287460	0.098549
	3.4	30.6560000	3.0648600	-3.0955200	0.3350700	0.099976
	3.5	30.6560000	3.0655800	-3.0962400	0.4344180	0.099999
	4	30.6560000	3.0656000	-3.0962500	0.9344020	0.100000
	5	30.6560000	3.0656000	-3.0962500	1.9344000	0.100000

Table #1: Common Collector Simulation Results

	I_{BQ} (µA)	I_{CQ} (mA)	V_{CEQ} (V)	DC Gain
$V_1 = 4V$	10.746	1.075	3.937	100
$V_1 = 6V$	17.372	1.737	3.263	100
$V_1 = 8V$	24.010	2.401	2.599	100
$V_1 = 10V$	30.656	3.066	1.934	100
Q_{mid}	20.791	2.070	2.936	100

Table #2: Common Collector Q-Point Behavior and Midpoint Biasing Q-Point